Kade Widler
Data Science & AI
BUas
7 April 2026

Automated Plant Phenotyping for Root System Architecture

Over the past four months, I built an end-to-end automated pipeline for measuring root system architecture in Arabidopsis thaliana seedlings grown in petri dishes. The system takes a raw image from the NPEC Hades phenotyping platform in Utrecht, segments the roots using a U-Net convolutional neural network, produces a root skeleton as inference, measures the primary root length, and maps detected root tip positions into physical coordinates, controlling a simulated robotic arm to navigate to each root tip and inoculate identified plants. The full pipeline spans computer vision, image processing, and robotics.

The most interesting part of this project was problem solving when building the pipeline. When I first ran inference on the test images, the results were full of errors. Water bubbles on the petri dish left arc-shaped artifacts in the segmentation masks. Detected noise as plants appeared in regions where no roots existed. The measurement algorithm followed lateral roots instead of the primary root, producing inflated length estimates.

Each of these problems were difficult to solve, but in each I spent way too long debugging, eventually arriving at simple solutions. The bubble artifacts had a distinctive elongated shape compared to actual roots, so I filtered connected components by their convex hull aspect ratio. The false detections outside plant regions disappeared when I defined vertical XY bands based on seed positions and discarded anything outside them. The lateral root measurement errors were addressed through heuristic pruning (I completed a research project on this testing smoothing and pruning methods of skeleton post-processing) detecting when the path travelled unnaturally upward (a signature of starting from a lateral root tip) and forcing the start point back to the crown of the plant. None of these solutions required retraining the model or adding complexity. They required looking carefully at the data, understanding the question, and building targeted fixes.

This poster presents the pipeline from raw image to robot simulation, with a focus on these three error analysis iterations. Each is shown as a problem-fix pair with before and after visuals. The theme, borrowed from Douglas Adams, is deliberate: DON'T PANIC. When faced with a broken pipeline and confusing outputs, the answer was right in front of me.

I chose to present this project because it reflects something that applies beyond AI and data science. When working on difficult problems, we tend to overcomplicate things. But in my experience across this project and the research that followed it, the most effective answers came from simple, interpretable methods grounded in framing. The pipeline now achieves a 3.2% sMAPE on the 19-image Kaggle test set, and each step is explainable to someone with no technical background, focusing on problem solving.